Sign-Balanced Liquidation Clearing in DeFi Markets

Abstract. Liquidations in DeFi lending protocols are interdependent: selling collateral moves prices, which can trigger or suppress other liquidations. We study a collateral-only liquidation model with linear price impact, in which liquidators repay debt externally and only collateral sales move market prices. Even in this conservative mechanism, clearing can be nonmonotone: when an asset sold as collateral is also used as another position's debt asset, its price decline reduces that borrower's debt burden.
We show that sign-balance has a precise complexity-theoretic meaning. If the asset borrowing graph is bipartite, then after flipping one side of the asset prices and complementing liquidation variables on one side, the smooth clearing equations become a subtraction-free monotone clipped-affine system. Approximate clearing in this sign-balanced regime reduces to the succinct Tarski fixed-point problem and therefore lies in CLS = PPAD $\cap$ PLS. In the small-impact regime, the same system is a contraction and clearing is computable in polynomial time. Conversely, once sign-balance is removed, non-bipartite instances can implement a NOT primitive, encode SUM/NOT generalized circuits, and finding even a constant-approximate clearing is PPAD-hard. Smooth approximate clearing is in PPAD. In the all-or-nothing model, the balanced case is polynomial-time solvable while connected non-bipartite clearing existence is NP-complete.

Keywords: DeFi liquidation · Clearing · PPAD · CLS · Tarski fixed points · Financial networks

1 Introduction and Technical Overview

DeFi lending protocols liquidate undercollateralized positions. A position has a collateral asset and a debt asset. If it becomes unhealthy, liquidators repay debt and receive collateral; at the market level, the relevant price impact is that collateral is sold. When positions are large relative to liquidity, these sales move prices, and those price movements affect the health of other positions. A clearing is a self-consistent vector of liquidations and prices: the liquidations implied by prices must be exactly the liquidations whose collateral sales generate those prices.

This paper studies the computational problem created by this feedback. We use a linear price-impact market as a first-order model of liquidation-induced price movement. This modeling choice is deliberately conservative. Liquidations only sell collateral; debts are repaid externally. There is no market buy pressure on the debt asset. Thus the nonmonotonicity we identify does not arise from an artificial assumption that liquidators buy debt in the same market. It arises from the fact that the same asset can appear as collateral in one position and as debt in another.

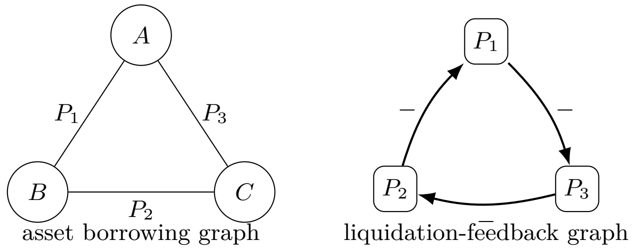

Fig. 1. A three-asset collateral-debt cycle. Selling an asset can lower another position's debt value, creating a negative liquidation dependency.

A DeFi-native negative feedback. Consider three assets A, B, C and three positions

$$P_1 = (A, B), \qquad P_2 = (B, C), \qquad P_3 = (C, A),$$

where (A, B) denotes collateral A and debt B. If P_1 is liquidated more aggressively, more A is sold and p_A falls. But A is the debt asset of P_3, so the value of P_3's debt falls. Thus P_3 becomes healthier and may liquidate less. A liquidation can therefore rescue another borrower whose debt is denominated in the asset being sold. This effect creates an all-negative cycle of liquidation feedback; see Fig. 1. It is the primitive behind both our sign-balance theorem and our hardness gadgets.

Positioning relative to monotone clearing. Financial clearing problems often have monotone or lattice structure. Eisenberg–Noe type payment networks are the classical example [14], and recent work makes the Tarski connection explicit: Besting, Hoefer, and Huth compute extremal Tarski fixed points in generalized financial networks with monotone piecewise-linear payment functions and default costs [1]. Fire-sale games also exhibit lattice equilibrium structure under convex price impact [2]. Our contribution is therefore not the observation that clearing can be phrased as a Tarski fixed point. The contribution is to identify what changes in DeFi liquidation clearing. The map here is not a conservation-based payment allocation; it is a price-impact fixed point in which liquidations change asset prices, and an asset sold as collateral can simultaneously be another borrower's debt asset. This debt-relief channel supplies a negative feedback primitive. We show that the collateral-debt graph determines whether this primitive can be globally aligned into a monotone Tarski/CLS problem or can instead implement SUM/NOT circuits and PPAD-hard fixed-point structure.

Main results. Let G_A be the asset borrowing graph: its vertices are assets and each position gives an edge between its collateral and debt assets. Our first main result identifies when the DeFi liquidation map preserves the Tarski-type structure familiar from monotone clearing.

> **Sign-balanced side.** If G_A is bipartite, smooth δ-clearing reduces to the succinct Tarski fixed-point problem and belongs to $\mathrm{CLS} = \mathrm{PPAD} \cap \mathrm{PLS}$.

If, in addition, liquidation feedback is small enough to make the transformed map contractive, clearing is unique and computable in polynomial time.

The proof uses the same sign transform as a monotonicity argument, but we do not claim that monotonicity alone gives a polynomial-time algorithm. After flipping one side of the asset prices and complementing liquidation variables whose collateral lies on one side, the equations become

$$y = \Pi_{[0,u]}(Ay + c), \qquad A \geq 0.$$

Thus the transformed map is order-preserving. Discretizing it on a grid yields a succinct monotone Tarski instance; a grid fixed point is an approximate clearing. The same representation also shows that the balanced equations have an upper-bounded Z-matrix complementarity structure. This structural reduction is useful, but by itself it is not a polynomial-time algorithm in the large-impact regime.

Our negative result gives the other side of the complexity map.

Non-bipartite worst case. There exists a constant $\delta_0 > 0$ such that, in the non-bipartite regime, finding a δ_0-approximate clearing is PPAD-hard.

We reduce from generalized circuits with only truncated-SUM and NOT gates, a restricted source problem known to be PPAD-complete for a constant approximation parameter. The reduction first ensures that the source circuit has an intrinsic odd-NOT cycle in the same dependency component as a hard-instance wire. The liquidation gadgets preserve this parity: NOT gates become odd collateral-debt paths, while SUM gates become even paths. Thus the hard instances are non-bipartite for the same reason that the source circuits are sign-frustrated. Finally, smooth approximate clearing belongs to PPAD. In the all-or-nothing model, the balanced case is polynomial-time solvable by finite monotone iteration, while the new discrete construction below gives NP-completeness for connected non-bipartite asset graphs.

The paper's contributions can be summarized as follows.

- *DeFi-native model:* a collateral-only liquidation clearing problem with linear price impact, close-factor-compatible sales, and debt relief through asset-denominated liabilities.
- *Sign-balance criterion:* bipartite asset borrowing graphs yield subtraction-free monotone clipped-affine fixed points, hence CLS membership for smooth approximate clearing.
- *Two-dimensional complexity map:* small feedback gain gives contraction and polynomial-time computation, while large-impact bipartite smooth clearing remains open.
- *Hardness primitive:* odd-NOT frustration in SUM/NOT circuits maps to non-bipartite collateral-debt cycles, giving constant-δ PPAD-hardness in the large-impact worst case.
- *Discrete hardness:* all-or-nothing clearing is NP-complete in connected non-bipartite instances, completing the discrete cell of the map.

Table 1. Complexity map. The hard entries are worst-case statements; small-impact entries are sufficient conditions for polynomial-time convergence.

	Sign-balanced / bipartite	Unbalanced / non-bipartite
Smooth	in CLS; small-impact P	constant-δ PPAD-hard
All-or-nothing	P	NP-complete

Relation to prior work. Prior work establishes Tarski/lattice structure, and in some settings polynomial algorithms, for conservation-like payment networks and strategic fire-sale models [1, 2]. We use these models as a foil. Our question is which DeFi network structures preserve Tarski-type monotonicity and which structures destroy it strongly enough to encode PPAD-hard fixed points. Credit-default-swap clearing hardness is another important antecedent [16, 4], but its nonmonotonicity comes from default-contingent derivative payments. Here, the NOT primitive comes from collateral-only liquidation lowering the price of another position's debt asset.

What the dichotomy does and does not say. The hardness theorem is a worst-case statement outside the balanced class. It does not say that every non-bipartite instance is hard. If price impact is sufficiently small, the clearing map may be a contraction even on a non-bipartite graph. The correct interpretation is structural: bipartiteness gives an order-preserving Tarski structure and CLS membership, while non-bipartite networks can encode PPAD-hard fixed-point structure when price impact is large enough.

2 Model and Sign Structure

There are assets $k \in [n]$, prices $p_k \in [0, M_k]$, and positions $i \in [m]$. Position i has collateral asset a_i, debt asset b_i, collateral amount $C_i > 0$, debt amount $D_i > 0$, liquidation threshold $\theta_i \geq 1$, and smoothing width $\varepsilon_i > 0$. We assume $a_i \neq b_i$. The health deficit is

$$h_i(p) = \theta_i D_i p_{b_i} - C_i p_{a_i}. \tag{1}$$

A positive deficit means the position is unhealthy. In the smooth model, liquidation intensity is

$$x_i = \Pi_{[0,1]}\left(\frac{h_i(p)}{\varepsilon_i}\right). \tag{2}$$

Liquidation sells collateral only. Thus the price equation is

$$p_k = \Pi_{[0,M_k]}\left(p_k^0 - s_k \sum_{i:a_i=k} \alpha_i x_i\right), \tag{3}$$

where $s_k > 0$ is the impact slope and $\alpha_i \geq 0$ is the collateral amount sold by a fully liquidated position. When we impose a close-factor constraint, we require

$0 \le \alpha_i \le \varphi C_i$ for a fixed $\varphi \in (0, 1]$. Debt is repaid externally and does not create a market buy order. The clipping makes the clearing map a continuous self-map of a compact box. The hardness construction below respects the close-factor bound; it adjusts output weights through market-depth parameters s_k, not by selling more collateral than a position contains.

Remark 1 (shared-collateral rank one). If two positions i, j share a collateral asset a, then their positive collateral-side interaction block has rank one:

$$\begin{pmatrix} C_i s_a \alpha_i & C_i s_a \alpha_j \\ C_j s_a \alpha_i & C_j s_a \alpha_j \end{pmatrix}, \qquad W_{ii} W_{jj} = W_{ij} W_{ji}.$$

Equivalently, all four positive couplings pass through the same price p_a and the same slope s_a. This constraint is used explicitly in the all-or-nothing appendix; debt-side negative reads are not tied by this identity because they can be scaled through separate debt assets and debt amounts.

A smooth exact clearing is a pair (p, x) satisfying the price and liquidation equations. A δ-clearing satisfies

$$\max\{\|p - P(x)\|_\infty, \|x - X(p)\|_\infty\} \le \delta,$$

where P and X denote the right-hand sides of (3) and the liquidation equation. In the all-or-nothing model, $x_i \in \{0, 1\}$; a clearing is a threshold-consistent liquidation set.

All parameters in the input are rational numbers encoded in binary. The computational problems ask for a δ-approximate clearing for a rational approximation parameter δ; in the contraction regimes below, exact or high-precision approximate clearings can be obtained by iteration. In the hardness theorem, δ is a fixed positive constant determined by the source generalized-circuit hardness constant and by the constant error amplification of the gadgets. The price clipping bounds M_k are part of the input; in the reduction they are constants.

We use two graphs. The asset borrowing graph G_A has assets as vertices and an undirected edge $\{a_i, b_i\}$ for each position. This is the graph used in the tractability and hardness statements. The position signed coupling graph G_X, formalized in Definition 1, has positions as vertices, a positive edge between i, j when $a_i = a_j$, and a negative edge when $a_i = b_j$ or $a_j = b_i$. For local intuition we also read it directionally: a positive arc $j \to i$ when $a_j = a_i$, and a negative arc $j \to i$ when $a_j = b_i$. Balance is a property of the undirected graph.

Economic meaning of bipartiteness. Bipartiteness should not be read as an arbitrary coloring condition. It says that assets can be separated into two sectors and all borrowing positions cross the sector boundary. A motivating special case is a market in which one sector contains volatile collateral-like assets and the other contains stable or debt-like assets, with positions of the form "deposit volatile, borrow stable" and "deposit stable, borrow volatile." The theorem does not require this interpretation, but it explains why bipartiteness is economically meaningful: it rules out odd cycles in which an asset returns with the opposite collateral/debt role, the source of the debt-relief negative feedback in Fig. 1.

Lemma 1 (price-level sign). *In the collateral-only model, the price map is competitive on the asset borrowing graph: if $m \neq k$, increasing p_m can only weakly decrease the next price of k, and only through positions with collateral k and debt m.*

Proof. For a position with collateral k and debt m, increasing p_m increases h_i, hence weakly increases x_i. This increases sales of asset k and weakly decreases the next price of k. Positions not using m as debt do not create this cross-effect. The clipping maps are nondecreasing, so the conclusion holds without differentiability assumptions.

At the liquidation-intensity level, if $a_j = a_i$, more liquidation of j depresses p_{a_i}, making i less healthy and increasing x_i: a positive dependency. If $a_j = b_i$, more liquidation of j depresses p_{b_i}, making i's debt cheaper and decreasing x_i: a negative dependency. Thus odd cycles of such negative dependencies are the obstruction to sign-monotonicity. In the directional reading of G_X, the three-asset example in Fig. 1 is an all-negative directed 3-cycle.

This diagnostic graph is useful for intuition, but the algorithmic boundary is stated on the asset graph G_A. In generic nondegenerate regions, a signed coordinate flip can make all direct liquidation dependencies monotone exactly when the signed feedback graph has no odd negative cycle, in the sense of Harary balance. The asset-level bipartite condition is the clean sufficient condition we use throughout the paper: every collateral-debt edge crosses the cut, so one global sign convention makes the competitive price interactions coherent.

If G_A is bipartite, a coordinated sign flip removes the competitive signs at the price level. More explicitly, the bipartition defines an orthant order on prices: prices on one side are ordered normally and prices on the other side are ordered in reverse. In that order the price map is isotone. A parallel transform on liquidation variables gives the sign-balanced order used below.

Sign balance versus hardness. The two graph views should not be conflated. The asset graph G_A controls the sign pattern of the price map, and bipartiteness of G_A is the structural assumption used on the sign-balanced side. The position graph G_X records local liquidation-to-liquidation feedback: co-writers with a common collateral asset produce positive edges, while a writer whose collateral is another position's debt asset produces a negative edge. Harary's balance criterion says that a signed graph admits a coordinate flip making all edges positive if and only if it has no cycle with an odd number of negative edges. The triangle in Fig. 1 violates this criterion.

The positive and negative results therefore have different quantifiers. Bipartiteness gives a monotone Tarski structure and places approximate clearing in CLS; it gives a polynomial-time algorithm under the additional small-impact condition of Proposition 2. Non-bipartiteness is a worst-case source of hardness: outside the balanced class, large-impact instances can be constructed that encode PPAD-complete fixed-point structure. Small-impact non-bipartite instances may still be contractions and hence easy. Sign balance is therefore a complexity-class certificate rather than an instance-level "easy if and only if" criterion.

The distinction between monotone structure and polynomial-time tractability is important. Tarski's theorem implies existence of fixed points and, in the succinct setting, a well-defined total search problem, but it does not by itself give a polynomial-time algorithm for a continuous map. The next section uses the monotone structure to place sign-balanced clearing in CLS, and then identifies a separate small-impact condition under which ordinary contraction gives polynomial-time computation.

3 Sign-Balanced Clearing, Tarski, and Small-Impact Tractability

Theorem 1. *In the collateral-only linear price-impact model with price clipping, if the asset borrowing graph G_A is bipartite, then for every rational $\delta > 0$, smooth δ-clearing reduces in polynomial time to the succinct Tarski fixed-point problem. Hence bipartite smooth δ-clearing belongs to* $\mathrm{CLS} = \mathrm{PPAD} \cap \mathrm{PLS}$.

Let (L, R) be a bipartition of G_A. Define transformed price variables

$$q_k = \begin{cases} p_k, & k \in L, \\ M_k - p_k, & k \in R, \end{cases} \tag{4}$$

and transformed liquidation variables

$$z_i = \begin{cases} 1 - x_i, & a_i \in L, \\ x_i, & a_i \in R. \end{cases} \tag{5}$$

Let $y = (q, z)$ and let $[0, u]$ be the corresponding box, with upper bounds M_k for transformed prices and 1 for transformed liquidations.

Lemma 2. *Under the above transform, the clearing equations are equivalent to*

$$y = \Pi_{[0,u]}(Ay + c) \tag{6}$$

for a rational matrix $A \geq 0$ and vector c. Moreover, after ordering price and liquidation variables separately, A has block form

$$A = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix}, \qquad P, Q \geq 0.$$

Proof. There are four cases. First, if $k \in L$, then every position with collateral k has $x_i = 1 - z_i$, and

$$q_k = p_k = \Pi_{[0,M_k]}\left(p_k^0 - \sum_{i:a_i=k} s_k \alpha_i + \sum_{i:a_i=k} s_k \alpha_i z_i\right),$$

which has nonnegative coefficients on the z_i's. Second, if $k \in R$, then $q_k = M_k - p_k$ and $x_i = z_i$ for positions collateralized by k. Using $M - \Pi_{[0,M]}(r) = \Pi_{[0,M]}(M - r)$,

$$q_k = \Pi_{[0,M_k]}\left(M_k - p_k^0 + \sum_{i:a_i=k} s_k \alpha_i z_i\right),$$

again with nonnegative coefficients.

For liquidations, consider $a_i \in L, b_i \in R$. Then $p_{a_i} = q_{a_i}$, $p_{b_i} = M_{b_i} - q_{b_i}$, and $z_i = 1 - x_i$. Since $1 - \Pi_{[0,1]}(r) = \Pi_{[0,1]}(1 - r)$,

$$z_i = \Pi_{[0,1]}\left(1 - \frac{\theta_i D_i M_{b_i}}{\varepsilon_i} + \frac{\theta_i D_i}{\varepsilon_i} q_{b_i} + \frac{C_i}{\varepsilon_i} q_{a_i}\right).$$

Finally, if $a_i \in R, b_i \in L$, then $z_i = x_i$, $p_{a_i} = M_{a_i} - q_{a_i}$, and $p_{b_i} = q_{b_i}$, giving

$$z_i = \Pi_{[0,1]}\left(-\frac{C_i M_{a_i}}{\varepsilon_i} + \frac{\theta_i D_i}{\varepsilon_i} q_{b_i} + \frac{C_i}{\varepsilon_i} q_{a_i}\right).$$

All displayed coefficients of variables are nonnegative. Price equations depend only on transformed liquidation variables, and liquidation equations depend only on transformed price variables, which gives the block-off-diagonal form.

Lemma 3 (gauge normal form). *Let $y_v = \Pi_{[0,1]}(y_u + y_w)$ be a truncated-SUM gate and write $\bar{t} = 1 - t$. Then*

$$\bar{y}_v = 1 - \Pi_{[0,1]}(y_u + y_w) = \max\{1 - y_u - y_w, 0\} = \Pi_{[0,1]}(\bar{y}_u + \bar{y}_w - 1).$$

Consequently, applying $y \mapsto 1 - y$ to a Harary-balanced subset of wires turns a SUM/NOT circuit into a subtraction-free clipped-affine system

$$y = \Pi_{[0,1]}(Ay + c), \qquad A \geq 0,$$

with nonnegative variable coefficients and a possibly signed constant vector c.

Proof. The displayed identity uses $1 - \Pi_{[0,1]}(t) = \Pi_{[0,1]}(1 - t)$ on the relevant range. A balanced signed dependency graph admits a vertex flip making every edge positive. Performing $y \mapsto 1 - y$ on the flipped wires replaces each NOT by an identity and each SUM by a gate of the displayed form, whose variable coefficients are nonnegative. Collecting gates gives the claimed clipped-affine system.

Corollary 1 (balanced circuit fragment). *The sign-balanced SUM/NOT fragment is the circuit analogue of the transformed DeFi system in Lemma 2: after a gauge transform it has the form $y = \Pi(Ay + c)$ with $A \geq 0$. Therefore the balanced fragment lies in CLS by Theorem 1; SUM/NOT hardness must come from sign-imbalance, i.e., odd-NOT frustration that cannot be gauged away.*

The map

$$T(y) = \Pi_{[0,u]}(Ay + c)$$

is therefore isotone in the product order. This is the true algorithmic consequence of sign-balance. It is stronger than a qualitative sign observation but weaker than a polynomial-time algorithm in the large-impact continuous setting: Tarski guarantees fixed points of monotone maps, and succinct monotone fixed-point search has its own complexity class behavior.

Proof (Proof of Theorem 1). Fix $\delta > 0$. Choose a rational grid step $\eta \leq \delta$ with polynomially many bits, and let

$$\mathcal{G}_\eta = [0, u] \cap \eta\mathbb{Z}^d$$

with the natural product order. Define a grid map

$$T_\eta(y) = \lfloor T(y) \rfloor_\eta,$$

where the floor is taken coordinate-wise to the largest grid point not exceeding $T(y)$. Since T is isotone and coordinate-wise rounding down is isotone, T_η is an order-preserving self-map of the finite grid lattice. It is also succinctly computable from the rational clearing instance.

A fixed point $y = T_\eta(y)$ is an η-approximate fixed point of T: coordinate-wise, $y \leq T(y) < y+\eta$, so $\|T(y)-y\|_\infty \leq \eta \leq \delta$. Transforming back gives a δ-clearing of the original market. Thus bipartite δ-clearing reduces to the succinct Tarski fixed-point problem. Etessami, Papadimitriou, Rubinstein, and Yannakakis show that the succinct Tarski problem lies in both PPAD and PLS [7]. Fearnley, Goldberg, Hollender, and Savani show that CLS = PPAD $\cap$ PLS [8]. Therefore bipartite δ-clearing belongs to CLS.

Theorem 2 (separable monotone price impact). *Replace the linear price equation by*

$$p_k = g_k\left(\sum_{i:a_i=k} \alpha_i x_i\right),$$

where each $g_k : [0, \infty) \to [0, M_k]$ is continuous, nonincreasing, and polynomial-time evaluable on rational inputs to any requested precision. If G_A is bipartite, then for every rational $\delta > 0$, smooth δ-clearing reduces in polynomial time to succinct Tarski and hence belongs to CLS. This includes independent asset–numeraire constant-product pools such as $g_k(Q) = \min\{\kappa_k/(R_k + Q), M_k\}$.

Proof. Use the same bipartition (L, R) and transformed variables q, z as in Theorem 1. The transformed map remains a continuous self-map of the same box. For the price block, if $k \in L$, then $x_i = 1 - z_i$ for positions collateralized by k, so the sale quantity $\sum_{i:a_i=k} \alpha_i x_i$ is nonincreasing in each z_i; since g_k is nonincreasing, $q_k = p_k$ is nondecreasing in each z_i. If $k \in R$, then $x_i = z_i$, the sale quantity is nondecreasing, $p_k = g_k(\cdot)$ is nonincreasing, and $q_k = M_k - p_k$ is nondecreasing.

For the liquidation block, the health equations are unchanged, so the four-case computation in Lemma 2 again gives transformed liquidation equations with nonnegative coefficients in the transformed prices. Hence the transformed map is isotone. Discretizing it on a rational grid and rounding down gives the same succinct Tarski reduction as in the proof of Theorem 1; polynomial-time evaluation of each g_k to grid precision suffices to compute the grid map.

Remark 2 (scope of the AMM extension). Theorem 2 is separable: each asset price depends only on the net amount of that asset sold. It covers independent single-pool or asset–numeraire monotone impact, but not routed, cross-asset, or path-dependent AMM execution. Those nonseparable AMM models remain outside the theorem.

Proposition 1 (sign-balanced comparative statics). *Fix a grid used in the proof of Theorem 1. Suppose two sign-balanced instances have transformed grid maps T_η and S_η with $T_\eta(y) \le S_\eta(y)$ for every grid point y. Then the least grid clearing of T_η is no larger than the least grid clearing of S_η, and the same monotone order holds for the greatest grid clearings.*

Proof. On a finite lattice, the least fixed point of an isotone map is obtained by iterating from the bottom element, and the greatest fixed point by iterating from the top. The pointwise inequality and isotonicity imply by induction that every bottom iteration under T_η is no larger than the corresponding bottom iteration under S_η. The top iteration gives the greatest fixed point comparison. In economic terms, parameter shocks that shift the transformed clearing map upward have ordered effects on the extremal sign-balanced clearings; in the original variables, the interpretation follows the same sign transform used in Theorem 1.

Proposition 2 (small-impact polynomial tractability). *In the transformed bipartite system $T(y) = \Pi_{[0,u]}(Ay + c)$, if $\rho(A) < 1$, then the clearing is unique and a δ-clearing can be computed in time polynomial in the input size and $\log(1/\delta)$.*

Proof. By Perron-Frobenius, since $A \ge 0$ and $\rho(A) < 1$, there are $v > 0$ and $\lambda < 1$ with $Av \le \lambda v$. In the weighted sup norm $\|r\|_v = \max_i |r_i|/v_i$, the projection $\Pi_{[0,u]}$ is nonexpansive coordinate-wise and

$$\|T(y) - T(y')\|_v \le \|A(y - y')\|_v \le \lambda \|y - y'\|_v.$$

Thus T is a contraction. Banach iteration from any point converges geometrically to the unique fixed point; with rational data, the standard bit-complexity bound for iterating a contraction gives a δ-clearing in time polynomial in the input and $\log(1/\delta)$.

Proposition 3 (Lipschitz tractability). *In the separable monotone-impact model of Theorem 2, suppose each g_k is γ_k-Lipschitz on the relevant sales range. Form the nonnegative gain matrix*

$$\widehat{A} = \begin{pmatrix} 0 & \widehat{P} \\ \widehat{Q} & 0 \end{pmatrix}, \qquad \begin{array}{l} \widehat{P}_{ki} = \gamma_k \alpha_i \quad (a_i = k), \\ \widehat{Q}_{ia_i} = \dfrac{C_i}{\varepsilon_i}, \qquad \widehat{Q}_{ib_i} = \dfrac{\theta_i D_i}{\varepsilon_i}. \end{array}$$

If $\rho(\widehat{A}) < 1$*, then the transformed clearing map is a contraction in a weighted sup norm, the clearing is unique, and a* δ*-clearing is computable in time polynomial in the input size and* $\log(1/\delta)$*.*

Proof. The clipping maps are coordinate-wise nonexpansive. The transformed price block is Lipschitz-bounded by $\widehat{P}$, while the transformed liquidation block is Lipschitz-bounded by $\widehat{Q}$, so

$$|T(y) - T(y')| \le \widehat{A}\,|y - y'|$$

component-wise. Since $\widehat{A} \ge 0$ and $\rho(\widehat{A}) < 1$, Perron-Frobenius gives $v > 0$ and $\lambda < 1$ with $\widehat{A}v \le \lambda v$. In the weighted norm $\|r\|_v = \max_i |r_i|/v_i$, the displayed bound gives $\|T(y) - T(y')\|_v \le \lambda \|y - y'\|_v$. Banach iteration yields uniqueness and geometric convergence; standard rational contraction bounds give the stated running time. For the linear model, $\gamma_k = s_k$, so Proposition 2 is recovered.

The transformed equations also have a useful complementarity interpretation. Let $B = I - A$ and $d = -c$. Then $y = T(y)$ is equivalent to the upper-bounded complementarity system

$$0 \le y \perp By + d + w \ge 0, \qquad 0 \le w \perp u - y \ge 0. \tag{7}$$

The base matrix B is a Z-matrix, since $A \ge 0$. The naive standard-LCP embedding of (7) has a $+I$ block and is not itself a Z-matrix LCP; the bounded form is the correct structural statement. Appendix A gives the detailed conversion. We use this complementarity view to explain the sign-balanced structure, but our polynomial-time claim is restricted to the contraction/M-matrix regime of Proposition 2.

Remark 3. The large-impact bipartite regime is an interesting open case. The transform places it in the Tarski/CLS world and gives the upper-bounded Z-matrix structure above, but we do not claim a polynomial-time algorithm for arbitrary $\rho(A) \ge 1$. This does not contradict polynomial algorithms for monotone payment-network clearing [1]. Those algorithms exploit payment-network structure such as no-overpayment/no-fraud constraints and conservation-like flow or circulation properties. The transformed DeFi map is monotone and piecewise-linear, but it is a price-impact liquidation map rather than a payment allocation map; its feedback matrix need not be substochastic or conservative, and in the large-impact regime it may have spectral radius at least one. In the small-impact regime where this feedback becomes contractive, Proposition 2 recovers polynomial-time computation.

4 Membership and All-or-Nothing Clearing

Proposition 4. *For rational inputs and rational* $\delta > 0$*, smooth* δ*-clearing belongs to PPAD.*

Proof. The right-hand sides of the price and liquidation equations are represented by a rational piecewise-linear circuit over addition, multiplication by rational constants, and min/max gates. The price clipping and liquidation clipping make this circuit a continuous self-map of a compact rational box. A δ-clearing is precisely an approximate fixed point of this piecewise-linear map, up to a harmless rescaling of residuals between price and liquidation coordinates. Approximate fixed points of rational piecewise-linear FIXP circuits are in PPAD via the Linear-FIXP/PPAD correspondence [9].

Proposition 5. *The all-or-nothing clearing existence problem belongs to NP. If the signed feedback structure is balanced, all-or-nothing clearing can be found in polynomial time.*

Proof. For NP membership, a certificate is the set of liquidated positions. Given this set, prices are computed directly from (3), and each threshold condition can be checked. In the balanced case, after the sign transform the induced map on $\{0, 1\}^m$ is monotone. Starting from the bottom element and iterating the map, the sequence is coordinate-wise nondecreasing in the transformed order. Since every coordinate is binary, each coordinate changes at most once, and the iteration stabilizes after at most m strict changes. This is the finite-lattice analogue of the monotone-clearing algorithms used in classical financial networks; unlike the smooth case, finiteness gives an immediate polynomial bound.

Theorem 3 (all-or-nothing non-bipartite hardness). *The all-or-nothing clearing-existence problem is NP-complete, even for collateral-only linear price-impact markets with rational parameters of polynomial bit complexity, strict threshold margins, bounded liquidation fractions $\alpha_i \leq \varphi C_i$, and connected non-bipartite asset borrowing graph G_A.*

Proof. Membership is Proposition 5: a liquidation set is a polynomial certificate, and prices and threshold inequalities can be checked directly. For hardness, Appendix B gives a polynomial reduction from 3-SAT. The construction uses free liquidation bits for variables, a DeFi-realizable threshold-gate library for the Boolean circuit, and a SAT-gated odd-NOT ring. If the formula is satisfied, the gate layer forces the controller on and the odd ring has a unique consistent state; if the formula is unsatisfied, the controller is off and the ring reduces to $r = \mathrm{NOT}(r)$, so no all-or-nothing clearing exists. The odd collateral-debt cycle is inside the same connected component as the SAT gate, and the construction respects the close-factor bound and the shared-collateral rank-one identities of Remark 1. Thus the bottom-right cell of Table 1 is NP-complete.

Construction roadmap. The appendix proves realizability in three layers. First, it isolates the rank-one constraint imposed by shared collateral and uses the corrected controller-ring witness $(w_{R_1}, U_W, w_W, u_W) = (1, 4, 4, 1)$. Second, it implements only a fixed threshold-gate library, not arbitrary signed weights. Third, a SAT-controlled odd-NOT ring makes the instance clear exactly when the formula is satisfiable and attaches the odd asset cycle to the Boolean component.

5 PPAD-Hardness in the Non-Bipartite Regime

Our hardness proof uses only two generalized-circuit gates. Let $T_{[0,1]}$ denote clipping to $[0,1]$. A κ-approximate assignment gives each wire a value in $[0,1]$ and makes every gate output differ from its prescribed value by at most κ. This is a local notion of satisfaction: circuits may contain cycles, and no topological evaluation order is assumed.

Lemma 4 (SUM/NOT source). *There is a constant $\kappa > 0$ such that finding a κ-approximate solution to generalized circuits with only the gates*

$$G_+(u,w) : y_v = T_{[0,1]}(y_u + y_w), \qquad G_{1-}(u) : y_v = 1 - y_u$$

is PPAD-complete, even with fan-out two.

We use the self-contained simplified generalized-circuit theorem of Budish, Gao, Othman, Rubinstein, and Zhang [6, Appendix D, Definition 10 and Remark 7] as the load-bearing source for this restricted gate set. Filos-Ratsikas et al. [5] give a related restricted-gate hardness result in first-price auction equilibrium computation. This source is important because truncated-SUM is already a primitive: our reduction does not build high-gain chains, and the error amplification below is independent of circuit size. In particular, the reduction uses neither comparison gates nor a band-reset construction; all gadgets below have constant self-feedback bounded away from one.

Lemma 5 (unbalanced SUM/NOT source). *Lemma 4 remains PPAD-hard under the promise that the signed dependency graph of the source circuit contains a cycle with an odd number of NOT gates in the same connected component as an original hard-instance wire.*

Proof. Given a SUM/NOT circuit C, pick any wire s in it. Add two fresh wires a, b and the two gates

$$a = 1 - s, \qquad b = T_{[0,1]}(s + a).$$

Every solution of the augmented circuit restricts to a solution of C, because the original gates are unchanged. Conversely, any exact solution of C extends by setting $a = 1 - s$ and $b = 1$. The new dependency graph contains the cycle $s - a - b - s$, with the edge $s - a$ coming from a NOT gate and the two edges incident to b coming from the SUM gate. Thus the cycle has odd NOT parity and lies in the connected component containing s. A κ-approximate solution of the augmented circuit gives a κ-approximate solution of C by projection, so a solver for this promised subclass would solve the PPAD-hard source problem.

Wire encoding and writer equation. For each circuit wire v, create an asset and encode $y_v = 2 - p_v$. Intended prices lie in $[1,2]$, inside the global clipping box $[0,3]$. A collateral-only writer with collateral output O, debt input U, and parameters C, D, ε satisfies

$$x = \Pi_{[0,1]}(c - a y_U + \lambda y_O), \qquad a = D/\varepsilon, \quad \lambda = C/\varepsilon, \quad c = (2D - 2C)/\varepsilon. \quad (8)$$

If this writer is the unique effective writer of O, then $y_O = x$. When $\lambda < 1$, the gadget defines a unique functional output and approximate-clearing residuals are amplified by at most a constant factor. All gadgets below have self coefficient at most $1/2$, and the NOT and averaging blocks use coefficient $1/3$. The construction never uses a high-gain amplification chain. The wire band is invariant: unique writers output $x \in [0,1]$, and the only co-writing operation is the self-bounding average gadget.

The construction respects a close-factor bound. If a writer should contribute η units of output price drop, take $C = 1$, $\alpha = \varphi/2$, and output slope $s_O = 2\eta/\varphi$. Then $s_O\alpha = \eta$ and $\alpha \leq \varphi C$. Thus coefficients are implemented through market depth, not by selling more collateral than a position contains.

The two gadgets. The NOT gate $y_O = 1 - y_U$ uses one writer with $C = 1, D = 2, \varepsilon = 3$, giving

$$y_O = \Pi_{[0,1]}\left(\frac{2 - 2y_U + y_O}{3}\right),$$

whose unique fixed point is $1 - y_U$. The truncated-SUM gate is a constant-size macro. First, two co-writers, each contributing half a unit of price drop and reading the complement wires $\bar{U}, \bar{W}$, create an average asset A satisfying $y_A = (y_U + y_W)/2$. Second, a NOT gadget creates $\bar{A}$. Finally, one writer with $C = 1, D = 2, \varepsilon = 2$ reads $\bar{A}$ and satisfies

$$y_O = \Pi_{[0,1]}\left(y_A + \frac{1}{2}y_O\right),$$

whose unique fixed point is $T_{[0,1]}(2y_A) = T_{[0,1]}(y_U + y_W)$. The algebra and constant error bounds are recorded in Appendix D. In particular, a source gate violation is at most $B\delta$ for an absolute constant B independent of the circuit size.

Lemma 6 (forward decoding). *For every SUM/NOT circuit C, one can construct in polynomial time a collateral-only liquidation instance I_C such that every δ-clearing of I_C decodes via $y_v = 2 - p_v$ to a $B\delta$-satisfying assignment of C, where $B = O(1)$ is independent of $|C|$.*

Proof. Replace each source gate by the corresponding constant-size macro. Debt-leg fan-out is non-invasive: an asset can be read as debt by many downstream writers without changing its price, since only collateral sales enter (3). Every public wire has a unique effective writer, except for the two intended co-writers inside the average macro. All self coefficients are bounded away from one, and every macro has constant size, so the error amplification is a universal constant. Generalized-circuit satisfaction is local over gates, hence no topological induction is needed even when the source circuit contains cycles.

The target search problem is total: clipping makes the clearing map continuous on a compact box, so Brouwer gives at least one exact clearing. A total-search reduction therefore needs only the forward decoding property above; we do not prove that every approximate circuit solution extends to a clearing.

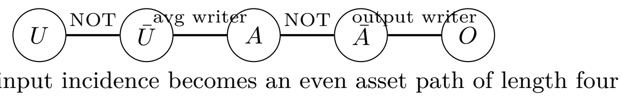

Fig. 2. Parity of a SUM macro. A NOT gate contributes an odd collateral-debt path, while each input-to-output incidence of a truncated-SUM macro is realized by the even path $U - \bar{U} - A - \bar{A} - O$.

Lemma 7 (gadget parity). *Let H_C be the signed undirected dependency graph of a SUM/NOT circuit C, with a negative edge for every NOT gate and positive edges from each input of a SUM gate to its output. For the liquidation instance I_C constructed above, with no additional disconnected anchor, the asset borrowing graph $G_A(I_C)$ is bipartite if and only if H_C is sign-balanced. In particular, an odd-NOT cycle in C maps to an odd cycle in $G_A(I_C)$.*

Proof. The claim is a parity accounting for the gadget templates; see Fig. 2. A NOT gate $U \to O$ is implemented by one writer whose collateral is O and debt is U, hence it contributes an odd path, in fact a single collateral-debt edge, between the two terminal assets. A SUM incidence $U \to O$ is implemented through the path

$$U - \bar{U} - A - \bar{A} - O,$$

where $U - \bar{U}$ and $A - \bar{A}$ are NOT-gadget edges and the other two edges are writer edges inside the averaging and final-output gadgets. This path has length four, hence even. The same holds for the other input $W \to O$. Internal paths inside a SUM macro between its terminals are even, and the internal subgraph has no odd cycle.

Thus the borrowing graph is obtained from the signed dependency graph by replacing negative edges by odd-length paths and positive edges by even-length paths, together with bipartite internal subdivisions. By Harary's parity criterion, such a graph is bipartite exactly when every cycle in H_C has an even number of negative edges. Therefore sign-balance of the source circuit is equivalent to bipartiteness of the constructed asset borrowing graph. The odd-NOT cycle supplied by Lemma 5 consequently gives an intrinsic non-bipartite cycle in the hard liquidation instance.

Constant approximation budget. Let $B_\neg$ and B_+ be the constant error amplifications of the NOT and SUM macros, and let B dominate both. These constants do not depend on the number of circuit gates. This independence is the main payoff from reducing from the restricted SUM/NOT source problem: the DeFi construction replaces each source gate by a constant-size macro and never simulates a comparator by a long amplification chain. If the clearing residual is at most δ, then each decoded source gate is violated by at most $B\delta$. Taking $\delta_0 = \kappa/(10B)$, where κ is the source approximation constant, converts any δ_0-clearing into a valid κ-solution of the source circuit.

Theorem 4. *There exists a constant $\delta_0 > 0$ such that, in the non-bipartite regime, finding a δ_0-approximate clearing in collateral-only linear price-impact liquidation markets is PPAD-hard. Consequently, for a sufficiently small fixed δ_0, the problem is PPAD-complete.*

Proof. Let κ be the constant from Lemma 5 and let B be the constant from Lemma 6. Choose $\delta_0 = \kappa/(10B)$. Given a promised unbalanced SUM/NOT circuit C, build I_C with no disconnected anchor. By Lemma 7, the asset borrowing graph of I_C is non-bipartite. Any δ_0-clearing decodes to a κ-satisfying assignment of C. Thus a clearing oracle for non-bipartite instances would solve a PPAD-complete source problem. Membership follows from the PPAD proposition.

The quantifiers in Theorem 4 are worst-case. We do not claim that every non-bipartite borrowing graph makes clearing hard. Rather, Lemma 7 shows that the constructed hard instances are non-bipartite for an intrinsic reason: the source circuit contains odd-NOT frustration, and the liquidation gadgets preserve that parity as an odd collateral-debt cycle. The non-bipartiteness is therefore part of the hard component, not a disconnected certificate attached after the fact.

6 Related Work and Discussion

Financial clearing as Tarski fixed points. Eisenberg and Noe [14] gave the classical monotone clearing model, and Rogers and Veraart [15] studied default costs. Recent work has made the Tarski viewpoint explicit. Besting, Hoefer, and Huth [1] compute extremal Tarski fixed points in generalized Eisenberg–Noe networks with monotone piecewise-linear payment functions and default costs, obtaining a strongly polynomial algorithm for minimal clearing and a polynomial algorithm for maximal clearing. This line is the right foil for our positive result. Their model remains a payment-network model with no-overpayment/no-fraud style constraints and conservation-like structure. Our transformed bipartite map is monotone and piecewise-linear, but it is generated by price impact and health-factor liquidation. It can have super-unit feedback gain and need not satisfy the substochastic or circulation structure used by payment-network algorithms. Thus the unconditional conclusion on the sign-balanced side is CLS/Tarski membership; polynomial-time computation follows under the separate small-impact condition.

Fire sales and lattice equilibria. Fire-sale models also connect price impact with lattice structure. Bertschinger et al. [2] study strategic fire-sale games and show that, for a broad range of price-impact functions such as convex impact, equilibria exist and form a complete lattice; they also analyze best-response convergence and cycling. Their setting is a strategic portfolio liquidation game. Ours is a DeFi lending clearing problem with collateral/debt roles, health-factor thresholds, and a graph-theoretic sign condition. We use Tarski structure only on the sign-balanced side, and we show that odd collateral-debt cycles can implement PPAD-hard SUM/NOT fixed-point structure.

CDS hardness and DeFi lending. Schuldenzucker, Seuken, and Battiston [16] showed PPAD-completeness of financial-network clearing with credit default swaps; Dohn, Hansen, and Klinkby [4] sharpened this hardness line and studied additional decision problems. These results show that derivative-based clearing has a rich PPAD/FIXP hardness theory. Our hardness uses no CDS, no default-contingent derivative payments, and no debt-side buy pressure. The NOT primitive is produced by collateral-only liquidation lowering the price of another position's debt asset. On the DeFi side, empirical and theoretical work documents liquidations, price impact, and systemic fragility [10–13]; Bartoletti and Lipparini [3] give a formal model of DeFi lending protocols and their incentive mechanisms. We complement this literature by isolating the fixed-point clearing problem created by price-impact liquidation feedback.

DeFi interpretation and limitations. The linear model is a first-order abstraction for aggregate liquidation pressure, and Theorem 2 shows that the sign-balanced CLS upper bound extends to separable monotone price impact, including independent asset–numeraire constant-product pools. Routed, cross-asset, and path-dependent AMM execution remains outside that theorem. Extending either the exact sign-balanced formulation or the gadget proof to those nonseparable AMM models is open. The hardness theorem is worst-case: small-impact non-bipartite instances may still be contractions and easy.

Open complexity of the balanced large-impact case. The large-impact sign-balanced regime is governed by the following abstract problem: given rational $A \geq 0$, c, and $u > 0$, find an exact or δ-approximate fixed point of

$$T(y) = \Pi_{[0,u]}(Ay + c).$$

We call this the *Subtraction-Free Box Fixed Point* problem. It is equivalent, via Lemma 3, to the balanced SUM/NOT fragment and, via Theorem 1, to the transformed sign-balanced DeFi clearing problem. Its box-LCP base is a Z-matrix; when $\rho(A) < 1$ it is an M-matrix/P-matrix case and Proposition 2 gives polynomial-time computation. For $\rho(A) \geq 1$, the upper bound is the obstruction: the naive standard-LCP embedding introduces a positive $+I$ block and leaves the Z-matrix world. Appendix F records a structural lemma and an EOPL-style attack route, but we leave the exact complexity of the large-impact sign-balanced case open.

Two partial results narrow this gap (Appendix G). Under γ-spectral separation of every active or residual SCC encountered by the recursive block algorithm, the *exact* least clearing is computable in polynomial time (Theorem 8). Under the bounded-magnitude assumptions listed in Corollary 2, inverse-polynomial-accuracy δ-clearing is also polynomial-time computable (Proposition 6). The unconditional large-impact case remains open; it reduces to an irreducible critical SCC ($\rho = 1$) with positive drift.

Conclusion. Collateral-only DeFi liquidation creates a native source of nonmonotone clearing feedback. Bipartite sign-balance gives a monotone Tarski instance

Table 2. Summary of the final frontier. The hard cells are worst-case; the small-impact guarantees are sufficient conditions.

	Sign-balanced / bipartite	Unbalanced / non-bipartite
Smooth	in CLS; small-impact P	constant-δ PPAD-hard
All-or-nothing	P	NP-complete

in CLS and a polynomial-time contraction problem under small impact; removing sign-balance enables SUM/NOT circuits and yields constant-approximation PPAD-hardness. In the discrete all-or-nothing model, the same frontier separates polynomial-time balanced clearing from connected non-bipartite NP-completeness.

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A Additional Details for Sign-Balanced Clearing

This appendix records the complementarity conversion used as a structural observation in Section 3. For $y = \mathit{\Pi}_{[0,u]}(Ay + c)$, define $B = I - A$ and $d = -c$. Then $F(y) = By + d$. Coordinate-wise, the fixed-point condition is equivalent to

$$y_i = 0 \Rightarrow F_i(y) \geq 0, \quad 0 < y_i < u_i \Rightarrow F_i(y) = 0, \quad y_i = u_i \Rightarrow F_i(y) \leq 0.$$

Introducing an upper-bound multiplier $w \geq 0$ gives the upper-bounded LCP formulation

$$0 \leq y \perp By + d + w \geq 0, \qquad 0 \leq w \perp u - y \geq 0.$$

Indeed, if $0 < y_i < u_i$, then $w_i = 0$ and $(By+d)_i = 0$. If $y_i = 0$ and $u_i > 0$, then $w_i = 0$ and $(By + d)_i \geq 0$. If $y_i = u_i$, then $(By + d)_i = -w_i \leq 0$. Coordinates with $u_i = 0$ may be eliminated in preprocessing.

The ordinary standard-LCP embedding of this upper-bounded system, with variables (y, w), has matrix

$$\begin{pmatrix} B & I \\ -I & 0 \end{pmatrix}.$$

The $+I$ block is forced by the sign of the upper-bound multiplier and gives positive off-diagonal entries. Hence the bounded system should not be described as an ordinary Z-matrix LCP. What bipartiteness gives unconditionally is the monotone clipped-affine form, the Tarski reduction, and the upper-bounded Z-matrix base structure. A polynomial-time algorithm follows in the small-impact regime where $\rho(A) < 1$, equivalently $I - A$ is a nonsingular M-matrix.

B All-or-Nothing NP-Hardness Construction

This appendix gives the construction used in Theorem 3. The order matters: first we record the realizability constraints of the liquidation model, then the fixed gate library, and only then the SAT-gated odd-NOT ring. In particular, we never claim that arbitrary signed threshold matrices are implementable.

B.1 Discrete Clearing as a Signed Threshold Network

Fix a collateral-only linear price-impact instance with positions $i \in [m]$, assets k, parameters $a_i, b_i, C_i, D_i, \theta_i, \alpha_i, p_k^0, s_k, M_k$, with $a_i \neq b_i$ and $0 \leq \alpha_i \leq \varphi C_i$. In the all-or-nothing model $x_i \in \{0, 1\}$ and

$$p_k = \mathit{\Pi}_{[0,M_k]}\left(p_k^0 - s_k \sum_{j:a_j=k} \alpha_j x_j\right).$$

When clipping does not bind, a clearing is a fixed point of the signed threshold update

$$x_i = \mathbf{1}[h_i(x) > 0], \qquad h_i(x) = \beta_i + \sum_j W_{ij} x_j, \tag{9}$$

where

$$\beta_i = \theta_i D_i p^0_{b_i} - C_i p^0_{a_i}, \qquad W_{ij} = \begin{cases} C_i s_{a_i} \alpha_j, & a_j = a_i, \\ -\theta_i D_i s_{b_i} \alpha_j, & a_j = b_i, \\ 0, & \text{otherwise.} \end{cases}$$

The positive terms are shared-collateral effects, and the negative terms are debt-relief reads. The self-coupling $W_{ii} = C_i s_{a_i} \alpha_i$ is positive.

Lemma 8 (rank-one positive blocks, restating Remark 1). *If positions i and j share a collateral asset a, then*

$$\begin{pmatrix} W_{ii} & W_{ij} \\ W_{ji} & W_{jj} \end{pmatrix} = \begin{pmatrix} C_i s_a \alpha_i & C_i s_a \alpha_j \\ C_j s_a \alpha_i & C_j s_a \alpha_j \end{pmatrix}$$

has rank one. In particular $W_{ii} W_{jj} = W_{ij} W_{ji}$. For the controller-ring pair below, this identity is

$$U_W u_W = w_{R_1} w_W.$$

Debt-read coefficients are not constrained by this identity.

Proof. The two columns are proportional with ratio α_j / α_i, so the determinant is zero. Debt reads go through the debt asset and can be scaled by $\theta_i D_i$ and the corresponding sale impact of the read asset, rather than through the shared collateral price.

Lemma 9 (band/no clipping). *For each asset k, let $\Sigma_k = \sum_{j:a_j=k} \alpha_j$. Set*

$$p^0_k = 2 s_k \Sigma_k + 1, \qquad M_k = p^0_k + 1.$$

Then for every $x \in \{0,1\}^m$, the price $p_k = p^0_k - s_k \sum_{j:a_j=k} \alpha_j x_j$ lies strictly inside $[0, M_k]$, so the outer clipping in (3) never binds. If the gadget biases are placed with margins at least one, the tie convention at $h_i = 0$ is irrelevant.

Proof. The price lies in $[p^0_k - s_k \Sigma_k, p^0_k] = [s_k \Sigma_k + 1, 2 s_k \Sigma_k + 1] \subset (0, M_k)$. The threshold margins are then checked in the affine system (9).

B.2 A Fixed DeFi-Realizable Gate Library

The construction uses dual rails: a signal u has an asset for u and a separate asset for $\bar{u}$. Free bits are bistable positions, and NOT gates create complementary rails.

Lemma 10 (fixed threshold gate library). *The following constant-size gates are realizable in the all-or-nothing DeFi model with fresh bus and output assets, polynomial-bit rational parameters, strict margins, and $\alpha_i \leq \varphi C_i$: a free bit, NOT, copy/literal, fan-in-k NOR via a fresh bus, the resulting OR/AND gates by dual rail, the SAT bit, and the gated ring of Lemma 13. Shared-collateral uses obey Lemma 8.*

Proof. A free bit P_v uses a unique collateral asset and an inert debt anchor, with $h_v = \beta_v + w_{vv}x_v$ and $-w_{vv} < \beta_v < 0$; both $x_v = 0$ and $x_v = 1$ are consistent. A NOT gate $Q = \mathrm{NOT}(P_v)$ uses debt asset σ_v and fresh collateral, with

$$h_Q = \beta_Q + w_Q x_Q - W_Q x_v,$$

and $0 < \beta_Q < W_Q - w_Q$, so the input swing dominates self-feedback.

For a NOR gate, route all inputs onto a fresh bus asset β_g. A copy-writer for input t sells the bus exactly when the corresponding input is true; the gate position reads the bus as debt and has a fresh private collateral asset. Its threshold is $h_g = \beta_g + w_g x_g - \sum_t c_t u_t$. Choose the bias in a gap of the attainable sums $\sum_t c_t u_t$ and choose w_g below the gap. The gate is then uniquely consistent and equals the intended downward threshold; OR and AND follow by dual rails and NOTs. Copy-writers sharing the bus obey the rank-one identity, while their faithfulness is maintained by scaling the relevant debt-read swings. The SAT bit is an acyclic composition of these fixed gates. All parameters use a fixed finite set of rational weights, except for polynomially many fresh copies, and the band choice in Lemma 9 prevents clipping.

Remark 4. The lemma is deliberately a fixed-library statement, not an arbitrary signed-matrix realizability theorem. This is the point of the rank-one discipline: only the gates actually used by the reduction are implemented.

B.3 The SAT-Gated Odd-NOT Ring

Build an odd ring with three positions

$$R_2 = (a_2; a_1), \qquad R_3 = (a_3; a_2), \qquad R_1 = (a_1; a_3),$$

so the ungated ring enforces $R_2 = \mathrm{NOT}(R_1)$, $R_3 = \mathrm{NOT}(R_2)$, and $R_1 = \mathrm{NOT}(R_3)$, i.e. $R_1 = \mathrm{NOT}(R_1)$. This has no consistent state and gives the odd asset cycle $a_1 - a_2 - a_3 - a_1$. A controller W shares collateral a_1 with R_1 and reads $\sigma_{\overline{\mathsf{SAT}}}$ as debt:

$$h_W = \beta_W + w_W x_W + u_W x_{R_1} - V_W x_{\overline{\mathsf{SAT}}},$$

$$h_{R_1} = \beta_{R_1} + w_{R_1} x_{R_1} - V_{R_1} x_{R_3} + U_W x_W.$$

The shared-collateral terms obey $U_W u_W = w_{R_1} w_W$. We require

$$\begin{aligned}
\mathsf{SAT} = 1: &\quad \beta_W > 0 \Rightarrow W = 1,\\
\mathsf{SAT} = 0: &\quad V_W > \beta_W + w_W + u_W \Rightarrow W = 0,\\
W = 0: &\quad 0 < \beta_{R_1} < V_{R_1} - w_{R_1} \quad \text{(clean NOT)},\\
W = 1: &\quad U_W > V_{R_1} - \beta_{R_1} \Rightarrow R_1 = 1.
\end{aligned}$$

Lemma 11 (rank-one-correct witness). *The domination inequalities and rank-one identity are simultaneously satisfied by*

$$(\beta_{R_1}, w_{R_1}, V_{R_1}, U_W) = (1, 1, 3, 4), \qquad (\beta_W, w_W, u_W, V_W) = (1, 4, 1, 7).$$

Equivalently, the shared-collateral tuple is $(w_{R_1}, U_W, w_W, u_W) = (1, 4, 4, 1)$.

Proof. Rank one holds because $U_W u_W = 4 \cdot 1 = 4 = 1 \cdot 4 = w_{R_1} w_W$. The ring inequalities give $0 < 1 < 3 - 1 = 2$ and $4 > 3 - 1 = 2$. The controller inequalities give $1 > 0$ and $7 > 1 + 4 + 1 = 6$.

Lemma 12 (close-factor realization). *The weights in Lemma 11 are realized with* $C_{R_1} = C_W = 1$*,*

$$\alpha_{R_1} = \frac{\varphi}{8}, \qquad \alpha_W = \frac{\varphi}{2}, \qquad s_{a_1} = \frac{8}{\varphi}.$$

Then $w_{R_1} = 1$*,* $U_W = 4$*,* $u_W = 1$*, and* $w_W = 4$*, while* $\alpha_{R_1}, \alpha_W \leq \varphi C$*. The debt reads* $V_{R_1} = 3$ *and* $V_W = 7$ *are realized independently through* a_3 *and* $\sigma_{\overline{\mathsf{SAT}}}$*, by scaling* θD *and the corresponding slopes. The biases are set by the* p^0*'s, and Lemma 9 keeps all prices in band.*

Lemma 13 (ring behavior). *Under Lemmas 11 and 12, the block* $\{W, R_1, R_2, R_3\}$ *has the unique consistent state*

$$(x_W, x_{R_1}, x_{R_2}, x_{R_3}) = (1, 1, 0, 1)$$

when $\mathsf{SAT} = 1$*. It has no consistent state when* $\mathsf{SAT} = 0$*.*

Proof. If $\mathsf{SAT} = 1$, then $\overline{\mathsf{SAT}} = 0$, so the controller inequality forces $x_W = 1$. The $W = 1$ inequality forces $x_{R_1} = 1$, and the clean NOTs force $x_{R_2} = 0$ and $x_{R_3} = 1$; rechecking W and R_1 gives consistency and uniqueness. If $\mathsf{SAT} = 0$, then $\overline{\mathsf{SAT}} = 1$, so $x_W = 0$. The ring then reduces to three clean NOTs and hence to $R_1 = \mathrm{NOT}(R_1)$, impossible.

B.4 Reduction from SAT

Given a 3-CNF formula ϕ over variables $v_1, \ldots, v_n$, build free bits $P_{v_1}, \ldots, P_{v_n}$, a NOT per variable for dual rails, an acyclic gate circuit computing

$$\mathsf{SAT} = \bigwedge_c (\ell_{c,1} \vee \ell_{c,2} \vee \ell_{c,3})$$

and $\overline{\mathsf{SAT}}$, and the gated ring above driven by $\overline{\mathsf{SAT}}$. All parameters are chosen according to Lemmas 9, 10, 11, and 12.

Theorem 5 (correctness of the discrete construction). *The constructed instance* I_ϕ *has an all-or-nothing clearing iff* ϕ *is satisfiable. Moreover, the number of clearings equals the number of satisfying assignments.*

Proof. By Lemma 9, all updates are affine threshold updates. Free bits are mutually independent and bistable. The literal, clause, SAT, and $\overline{\mathsf{SAT}}$ layers form an acyclic functional circuit over the free bits by Lemma 10. The gated ring is a sink: it is read by no upstream gate. Thus a global fixed point is exactly a choice of free-bit assignment, its unique functional completion, and a consistent ring state. By Lemma 13, the ring has a consistent state exactly when $\mathsf{SAT} = 1$, and then that state is unique. This gives the bijection between clearings and satisfying assignments.

Lemma 14 (connected non-bipartite asset graph). *The asset borrowing graph $G_A(I_\phi)$ contains the odd triangle $a_1 - a_2 - a_3 - a_1$, and the controller edge $\{a_1, \sigma_{\overline{\mathsf{SAT}}}\}$ attaches this triangle to the component carrying the SAT circuit and variables. Hence the relevant component is connected and non-bipartite.*

Proof. The three NOTs in the ring give the edges $\{a_1, a_2\}$, $\{a_2, a_3\}$, and $\{a_3, a_1\}$. The controller $W = (a_1; \sigma_{\overline{\mathsf{SAT}}})$ adds $\{a_1, \sigma_{\overline{\mathsf{SAT}}}\}$. The SAT circuit is connected to $\sigma_{\overline{\mathsf{SAT}}}$ by construction. Adding edges to a graph containing an odd cycle preserves non-bipartiteness.

Theorem 6 (Theorem 3, restated). *All-or-nothing clearing existence is NP-complete under the restrictions stated in Theorem 3. If the signed coupling graph is balanced, a clearing can be found in polynomial time as in Proposition 5.*

Proof. Membership is Proposition 5. For hardness, the construction has size polynomial in $|\phi|$, rational coefficients of polynomial bit complexity, strict margins, and $\alpha_i \leq \varphi C_i$. Theorem 5 shows that I_ϕ clears iff ϕ is satisfiable, and Lemma 14 shows that the hard instances are connected and non-bipartite. This is a polynomial many-one reduction from 3-SAT.

Remark 5 (computational sanity check). We additionally validated the gadget truth tables and the rank-one realizability constraint with brute-force fixed-point enumerators on small instances. The proof above rests on the inequalities; the checkers are only a sanity-check artifact.

C Discrete Parity Ledger

This optional structural appendix mirrors the smooth SUM/NOT parity lemma. It is not needed for Theorem 3; its role is to show that in the discrete construction, imbalance is localized to the gated odd-NOT ring.

Definition 1 (signed coupling graph). *This is the undirected symmetrization of the diagnostic position graph from Section 2. The vertices of G_X are positions. For $i \neq j$, add a positive edge when $a_i = a_j$ and a negative edge when $a_i = b_j$ or $a_j = b_i$. Sharing only a debt asset induces no edge.*

Lemma 15 (bipartite implies balanced). *If G_A is bipartite, then G_X is balanced.*

Proof. Let χ be a two-coloring of G_A, and set $\sigma_i = (-1)^{\chi(a_i)}$. If $a_i = a_j$, then $\sigma_i \sigma_j = +1$. If $a_j = b_i$ or $a_i = b_j$, then the collateral and debt assets of the corresponding position have opposite colors, so $\sigma_i \sigma_j = -1$. Thus edge signs equal $\sigma_i \sigma_j$.

Lemma 16 (converse under unique effective writers). *Suppose every priced asset is the collateral of exactly one effective writer. Suppose also that all other debt anchors are private pendant assets. If G_X is balanced, then G_A is bipartite.*

Proof. Let σ_i be a balancing gauge. For each priced asset k, color it by the sign of its unique writer: $(-1)^{\chi(k)} = \sigma_{w(k)}$. For a position i, its collateral a_i has writer i, so $(-1)^{\chi(a_i)} = \sigma_i$. If b_i is priced with writer j, then $a_j = b_i$, so $\{i, j\}$ is a negative edge and balance gives $\sigma_j = -\sigma_i$; hence a_i and b_i receive opposite colors. If b_i is an inert private anchor, color it opposite a_i. Thus every edge of G_A crosses the cut.

Theorem 7 (localized frustration). *In I_ϕ, deleting the three ring positions R_1, R_2, R_3 leaves a balanced signed coupling graph and a bipartite asset graph. The imbalance is localized to the gated-ring block: R_1, R_2, R_3 form an odd-NOT triangle, while the controller W has the side-couplings needed to gate R_1 and attaches that block to the SAT circuit. In $G_A(I_\phi)$, W contributes only the asset edge $\{a_1, \sigma_{\overline{\mathsf{SAT}}}\}$; the triangle $a_1 - a_2 - a_3 - a_1$ is the unique odd cycle of the ring block, and $\{a_1, \sigma_{\overline{\mathsf{SAT}}}\}$ connects it to the feed-forward component.*

Proof. The feed-forward gate library is acyclic in the Boolean sense and can be assigned the layer-parity gauge: cross-layer debt-read edges are negative and same-bus shared-collateral edges are positive. Lemma 15 gives the matching bipartite coloring of its asset graph. In G_X, the ring contributes the three negative edges R_1R_2, R_2R_3, R_3R_1, producing a negative 3-cycle. The controller contributes a positive shared-collateral edge WR_1, the negative side-coupling WR_2 because W's collateral asset a_1 is R_2's debt asset, and a negative debt-read attachment to the writer of $\sigma_{\overline{\mathsf{SAT}}}$ in the SAT component. Removing R_1, R_2, R_3 removes the ring obstruction and the WR_1, WR_2 side-couplings; the remaining controller attachment is part of the feed-forward balanced component. In G_A, the controller contributes only $\{a_1, \sigma_{\overline{\mathsf{SAT}}}\}$, which attaches the ring asset triangle to the feed-forward component without adding a second odd cycle inside the ring block.

D Gadget Algebra and Error Bounds

We use the encoding $y_v = 2 - p_v$. For a writer with collateral O, debt U, and parameters C, D, ε, the equation is

$$x = \Pi\left(\frac{2D - 2C}{\varepsilon} - \frac{D}{\varepsilon} y_U + \frac{C}{\varepsilon} y_O\right).$$

If it is the unique writer of O, then $y_O = x$. If $C/\varepsilon < 1$, the right-hand side is a contraction in y_O, and any residual δ gives an output error at most $\delta/(1 - C/\varepsilon)$ up to the harmless constant accounting for price and liquidation residuals.

For the NOT gadget, $C = 1, D = 2, \varepsilon = 3$. Hence

$$y_O = \Pi\left(\frac{2 - 2y_U + y_O}{3}\right),$$

whose unique fixed point is $1 - y_U$. The self coefficient is $1/3$, so a conservative error bound is $B_\neg = 3$.

For the average gadget, first maintain complement wires $\bar{U}, \bar{W}$ by NOT gadgets. Create an average asset A with two co-writers, both with collateral A, each contributing $\eta = 1/2$ units of output price drop. Writer 1 reads $\bar{U}$, writer 2 reads $\bar{W}$, and both use $C = 1, D = 2, \varepsilon = 3$. Since $y_{\bar{U}} = 1 - y_U$, writer 1 satisfies

$$x_1 = \frac{2}{3} y_U + \frac{1}{3} y_A,$$

and similarly $x_2 = \frac{2}{3} y_W + \frac{1}{3} y_A$. The output equation is $y_A = (x_1 + x_2)/2$, so $y_A = (y_U + y_W)/2$. The self coefficient is again bounded away from one, and the error is a constant multiple of δ.

The final gain-2 clipped writer reads $\bar{A}$ and uses $C = 1, D = 2, \varepsilon = 2$. Since $p_{\bar{A}} = 2 - (1 - y_A) = 1 + y_A$, the writer equation becomes

$$y_O = \Pi \left(y_A + \frac{1}{2} y_O \right).$$

If $y_A \leq 1/2$, the unique fixed point is $2y_A$; if $y_A > 1/2$, the unique fixed point is 1. Thus the output is $T_{[0,1]}(2y_A) = T_{[0,1]}(y_U + y_W)$. All self coefficients are at most 1/2 or 1/3 inside the macro, so the total error bound B_+ is an absolute constant.

The close-factor bound can be met by scaling the output impact. For a writer with desired output contribution η, take collateral amount parameter $C = 1$, liquidation amount $\alpha = \varphi/2$, and slope $s_O = 2\eta/\varphi$. Then $s_O \alpha = \eta$ and $\alpha \leq \varphi C$.

E Source Problem and Fan-Out

The restricted source problem in Lemma 4 is taken as a black box. The load-bearing citation is the full version of Budish et al. [6, Appendix D, Definition 10 and Remark 7], which gives a self-contained reduction to simplified circuits containing only SUM and NOT gates, even with bounded fan-out. Filos-Ratsikas et al. [5] and Rubinstein [17] provide related restricted-gate and generalized-circuit hardness background. Our construction does not require bounded fan-out because debt-leg reads are non-invasive.

The unbalanced-source promise used in Lemma 5 is not a target-side graph tag. It is obtained before translation by adding the two gates $a = 1 - s$ and $b = T_{[0,1]}(s + a)$ through an original source wire s. This creates a signed source cycle $s - a - b - s$ with one NOT edge. Lemma 7 then maps that source frustration to an odd collateral-debt cycle in the constructed borrowing graph.

F Subtraction-Free Structure of the Balanced Side

The sign-balanced map in Theorem 1 has the special form

$$y = \Pi_{[0,u]}(Ay + c), \qquad A \geq 0.$$

It is therefore monotone and subtraction-free: it can form nonnegative weighted sums and clip against constants, but it has no direct primitive for comparing two variables. Boolean monotone fragments such as AND/OR have easy least fixed points by finite Kleene iteration, while large-domain Tarski hardness constructions typically use min/max or other variable-comparison operations. Implementing such operations appears to require an anti-monotone dependence, which is precisely what the bipartite sign transform rules out.

LCP location. Writing $B = I - A$, the fixed-point condition is the box complementarity system

$$\begin{aligned} y_i = 0 &\Rightarrow (By + d)_i \geq 0, \\ 0 < y_i < u_i &\Rightarrow (By + d)_i = 0, \\ y_i = u_i &\Rightarrow (By + d)_i \leq 0, \qquad d = -c. \end{aligned}$$

Because $A \geq 0$, B is always a Z-matrix, and the equivalences

$$\begin{aligned} B \text{ is a } P\text{-matrix} &\iff B \text{ is a nonsingular } M\text{-matrix} \\ &\iff \rho(A) < 1 \end{aligned}$$

locate the regimes: small impact is an M-matrix box LCP, while large impact has a Z-matrix base that need not be a P-matrix. The unbounded Z-matrix LCP is polynomial for every Z-matrix; the obstruction here is the upper bound, whose standard-LCP embedding introduces the positive block shown in Appendix A.

Lemma 17 (extremal interior blocks are non-expansive). *Let y^* be the least fixed point of $T(y) = \Pi_{[0,u]}(Ay + c)$ with $A \geq 0$, and let $S = \{i : 0 < y_i^* < u_i\}$. Then $\rho(A_{SS}) \leq 1$. The same holds, by the dual argument, for the interior block of the greatest fixed point.*

Proof. Suppose $\rho(A_{SS}) > 1$. Since $A_{SS} \geq 0$, Perron–Frobenius gives an eigenvector $v \geq 0$, $v \neq 0$, with $A_{SS}v = \lambda v$ and $\lambda > 1$. Define $\tilde{y}_S = y_S^* - \varepsilon v$ and $\tilde{y}_{\bar{S}} = y_{\bar{S}}^*$. For $\varepsilon > 0$ small enough, $\tilde{y}_S \in (0, u_S)$, so the box projection is inactive on S and

$$T_S(\tilde{y}) = (A\tilde{y} + c)_S = y_S^* - \varepsilon A_{SS} v = y_S^* - \varepsilon \lambda v \leq \tilde{y}_S.$$

For a lower-boundary coordinate i, monotonicity and $\tilde{y} \leq y^*$ give $T_i(\tilde{y}) \leq T_i(y^*) = 0 = \tilde{y}_i$. For an upper-boundary coordinate, projection gives $T_i(\tilde{y}) \leq u_i = \tilde{y}_i$. Hence $T(\tilde{y}) \leq \tilde{y}$, so $\tilde{y}$ is a prefixed point, while $\tilde{y} < y^*$. By Knaster–Tarski, the least fixed point is below every prefixed point, contradiction. The greatest-fixed-point statement follows by pushing upward by εv and using the dual postfixed-point characterization.

Why exact active-set methods are delicate. Along the bottom Kleene sequence each coordinate has a monotone phase structure: it can move only from the lower boundary toward the interior and from the interior toward the upper boundary, never in the opposite direction. We do not assert a finite phase-transition bound, since a coordinate may approach its upper bound only asymptotically. Within a fixed active set the dynamics is an affine recursion $y_S^{t+1} = A_{SS} y_S^t + g$, and exact event-driven algorithms appear delicate because first-threshold times reduce to sign questions for matrix-power sequences. Moreover, for $\rho(A) \ge 1$ the box LCP can have several solutions, obstructing the unique-solution structure of UEOPL.

An EOPL-style attack route. The monotone grid iteration $y^{t+1} = T_\eta(y^t)$ with potential $\phi(y) = \mathbf{1}^\top y/\eta$ is a natural path-following viewpoint: T_η is isotone and rounds down, so the potential strictly increases off fixed points and is polynomially bounded. Turning this into an EOPL reduction would require a polynomial predecessor selection for the monotone Kleene path; the forward map is single-valued while predecessors need not be. We therefore record this only as an attack route. If it succeeds, the sign-balanced large-impact case would descend from CLS to an EOPL-style subclass.

Open problem. We leave open the exact complexity of large-impact sign-balanced clearing, equivalently of the Subtraction-Free Box Fixed Point problem, equivalently of the balanced SUM/NOT fragment: whether it is in P, in an EOPL-style subclass, or strictly between, for both exact and δ-approximate clearing. Appendix G records conditional and approximate polynomial-time algorithms and isolates the remaining $\rho = 1$ critical subcase.

G Subtraction-Free Box Fixed Points: An Algorithmic Refinement

We refine the balanced map $T(y) = \Pi_{[0,u]}(Ay + c)$, $A \ge 0$, of Theorem 1 and Appendices A and F, isolating the single source of difficulty in the large-impact ($\rho(A) \ge 1$) regime. Let y^* be its least fixed point, L the input bit length, and $U = \max_i u_i$. A point $y \in [0, u]$ is a δ-*approximate fixed point* if $\|T(y) - y\|_\infty \le \delta$, the residual notion in which δ-clearing is posed. Throughout we use the interior spectral bound: at y^*, the strictly interior block $S^* = \{i : 0 < y_i^* < u_i\}$ satisfies $\rho(A_{S^*S^*}) \le 1$ by Lemma 17.

SCC-DAG decomposition. Form the dependency digraph with arc $i \to j$ iff $A_{ij} \ne 0$. Let $C_1, \dots, C_m$ be its strongly connected components, ordered so that whenever C_b reads C_a we have $a < b$.

Lemma 18 (topological resolution). *$y^*_{C_a}$ is the least fixed point of*

$$T_a(z) = \Pi_{[0,u_{C_a}]}(A_{C_aC_a} z + g_a), \qquad g_a = c_{C_a} + \sum_{b<a} A_{C_aC_b} y^*_{C_b},$$

where g_a is fixed by upstream blocks.

Proof. C_a reads only C_a and components it depends on, among $C_1, \dots, C_{a-1}$. Thus $y^*_{C_a}$ is a fixed point of T_a. Minimality follows by induction, since the global bottom Kleene orbit restricted to C_a is the orbit of T_a once upstream has converged.

A Jordan block of A at the unit circle can arise only from coupling *between* critical SCCs, and is linearized away by resolving each upstream critical SCC to its constant limit. Within each C_a, the matrix $A_{C_a C_a}$ is irreducible, so its peripheral spectrum is ρ times roots of unity and semisimple.

Lemma 19 (residual locality). *For any $y \in [0, u]$ assembled block by block in topological order,*

$$\|T(y) - y\|_\infty = \max_a \|T_a(y_{C_a}) - y_{C_a}\|_\infty,$$

with each T_a using the stored upstream values. Hence resolving every block's sub-map to residual at most δ yields global residual at most δ.

Proof. For $i \in C_a$, $(Ay + c)_i = (A_{C_a C_a} y_{C_a} + g_a)_i$ with g_a formed from the stored upstream values, so $T(y)_i = T_a(y_{C_a})_i$.

Remark 6. Lemma 19 concerns the residual $\|T(y) - y\|$, the clearing accuracy of interest. Approximating y^* in value would require a Lipschitz propagation bound and per-block accuracy depending on downstream gains; we do not need it.

Per-block taxonomy. It suffices to solve one irreducible block $z = \Pi_{[0,u_S]}(Bz + g)$, $B = A_{SS} \geq 0$, with $r = \rho(B)$.

Lemma 20 (LP detects divergence). *For any $B \geq 0$, the polyhedron $P = \{z \geq 0 : (I - B)z \geq g\}$ is closed under coordinate-wise minimum. If $P \neq \emptyset$, it has a least element, returned by minimizing $\mathbf{1}^\top z$ over P. Moreover, $P = \emptyset$ iff the lower-bounded relaxation $z = \max(Bz + g, 0)$ diverges to $+\infty$.*

Lemma 21 (subcritical and LP-bounded blocks are exact). *If the least element $z^\star$ of P exists with $z^\star \leq u_S$, in particular when $r < 1$ or when $r = 1$ with nonpositive critical drift, then $z^\star$ is the exact box fixed point, computed in $\operatorname{poly}(|S|, L)$ time by the LP of Lemma 20. If $z^\star \not\leq u_S$, the coordinates with $z_i^\star > u_i$ are clamped to u_i and the reduced system is re-solved, for at most $|S|$ rounds.*

Proof. $z^\star$ solves the lower-bounded relaxation, whose map dominates T on $[0, u_S]$. Hence the box least fixed point is at most $z^\star$. If $z^\star \leq u_S$, the box projection is inactive at $z^\star$, so $z^\star$ is itself a box fixed point and therefore equals the box least fixed point. For $r < 1$, $I - B$ is a nonsingular M-matrix and the LCP has a unique least solution, found in polynomial time.

Remark 7 (relaxation is an upper bound). P drops the box, so its least element dominates y_S^*. Thus "the relaxation diverges" does not imply that the box system diverges to u: a supercritical block can settle finitely when a coordinate clips to 0, breaking the feedback. This is why each block is resolved inside its own box.

Lemma 22 (ramp step counts). *For an irreducible block with $r \geq 1$ whose lower-bounded LP is infeasible, bottom Kleene reaches its first clamp event in $O(U/\bar{\alpha}) \leq O(U/\delta)$ iterations when $r = 1$, where $\bar{\alpha}$ is the asymptotic per-step drift envelope over the peripheral period, unless $\bar{\alpha} \leq \delta$, in which case the current iterate is already a δ-approximate fixed point; and in $O(\log U/\log r)$ iterations when $r > 1$. Each block contributes at most $|S|$ clamp events.*

Proof. The interior increment is $d^{(t)} = B^t g$. For $r = 1$, the semisimple peripheral spectrum gives linear growth of the partial sums along the Perron direction, with a bounded periodic fluctuation when the block is imprimitive. Taking $\bar{\alpha}$ as the maximum drift over one peripheral period, the position advances at rate at most a constant-period oscillation around $t\bar{\alpha}\hat{v}$, so a cap is reached after $O(U/\bar{\alpha})$ steps; if this envelope is at most δ, the residual is already at most δ. For $r > 1$, the Perron mode dominates and the cap U is reached after $O(\log U/\log r)$ steps.

Approximation, scoped.

Proposition 6 (δ-approximate least fixed point). *Resolving blocks in topological order, using Lemma 21 for subcritical or LP-bounded blocks and bottom Kleene with Lemma 22 for ramping blocks, returns a δ-approximate fixed point in time*

$$O\big(n^2\,(nU/\delta + n\log U/\log r_{\min})\big) = \mathrm{poly}(n, L, \|A\|_\infty, U, 1/\delta).$$

Consequently, for normalized, polynomially bounded instances with $U, \|A\|_\infty = \mathrm{poly}(n, L)$ and $\delta \geq 1/\mathrm{poly}(n, L)$, the running time is polynomial. For arbitrary $A \geq 0, c, u, \delta$, the algorithm is pseudo-polynomial: polynomial in $n, L, \|A\|_\infty, U, 1/\delta$.

Proof. By Lemma 19, the global residual is the maximum per-block residual. Subcritical or LP-bounded blocks have residual 0. A ramping block either has increment $\bar{\alpha} \leq \delta$ or reaches a clamp within the count of Lemma 22, after which it is re-decomposed. Iterating leaves every block with residual at most δ. The time bound follows from at most $|S|$ clamps per block over at most n blocks, each preceded by $O(U/\delta)$ or $O(\log U/\log r)$ iterations of cost $O(n^2)$.

Corollary 2 (approximate clearing for balanced large-impact markets). *In the transformed system of Lemma 2, $y = (q, z)$ has box upper bounds M_k on prices and 1 on liquidations, and A has off-diagonal entries $s_k\alpha_i$ in the price block and $\theta_i D_i/\varepsilon_i, C_i/\varepsilon_i$ in the liquidation block. Hence $U = \max_k M_k$ and $\|A\|_\infty$ may be exponentially large binary rationals. For instances in which the price caps M_k, impact products $s_k\alpha_i$, and inverse smoothing widths $\theta_i D_i/\varepsilon_i, C_i/\varepsilon_i$ are polynomially bounded, a δ-clearing is computable in polynomial time for inverse-polynomial δ. In general the algorithm is pseudo-polynomial in these magnitudes and $1/\delta$.*

Exact solution under spectral separation.

Definition 2 **(γ-spectral separation).** *An instance is* γ-spectrally separated *if every irreducible block $B = A_{SS}$ that arises during the full recursive SCC-DAG resolution, including every residual subblock created after clamping coordinates to* 0 *or u, satisfies $\rho(B) \leq 1 - \gamma$ or $\rho(B) \geq 1 + \gamma$, unless the lower-bounded relaxation LP for B is feasible with least element at most u_S, so the block is resolved exactly by the LP with no ramp.*

Theorem 8 **(exact polynomial-time solution).** *If an instance is γ-spectrally separated with $\gamma \geq 1/\mathrm{poly}(n, L)$, then the exact least fixed point y^* is computable in* $\mathrm{poly}(n, L)$ *time, even when the box bounds are exponentially large. No tie-breaking hypothesis is needed.*

Proof. Every block encountered by the recursive algorithm is subcritical, LP-bounded, or supercritical with $r \geq 1 + \gamma$; none is a critical or barely-supercritical ramp. Subcritical and LP-bounded blocks are exact in polynomial time. A supercritical block reaches its next clamp, or is driven to 0 by lower clipping, within $O(\log U/\log(1+\gamma)) = O(L/\gamma) = \mathrm{poly}(n, L)$ bottom-Kleene iterations. After the clamp, the algorithm re-decomposes the residual subblocks, which are covered again by Definition 2. Thus each of the at most $|S|$ clamp events in the recursive resolution has the same polynomial bound. Once no clamp remains, the interior residual is obtained by one exact linear solve. Simultaneous boundary hits are handled by the clip, so no separation of hitting times is needed.

The critical event-jump. The sole obstruction to bit-length polynomiality in Proposition 6 and to dropping separation in Theorem 8 is one subcase: an irreducible block with $\rho = 1$ and positive drift ramping to its box.

Proposition 7 **(null-vector ramp event, proven special case).** *Let $B \geq 0$ be irreducible with $\rho(B) = 1$, constant input g, and suppose the block ramps, i.e., the LP of Lemma 20 is infeasible. Let $v > 0$ be the positive right null vector of $B - I$. Then the bottom Kleene orbit advances along v, and, when B is primitive or the headroom ratios are separated by $\min_{i \neq j} |u_i/v_i - u_j/v_j| \geq 2^{-\mathrm{poly}(n,L)}$, the first coordinate to reach its cap is $j = \arg\min_i u_i/v_i$, independently of the drift magnitude. Clamping j and recursing terminates in at most $|S|$ events, each a null-vector plus a ratio test, in* $\mathrm{poly}(|S|, L)$ *time with no dependence on U or the drift.*

Status. An unconditional event-jump still requires period-h phase alignment for near-tie boundary events, rational bit-complexity of the null-vector and ratio computations, and the post-clamp residual when the reduced block is supercritical. These remain empirically passing but not used to support any theorem above.

Refinement of the complexity map.

Regime (smooth model)	**Result**
sign-balanced	in CLS
sign-balanced + small impact ($\rho(A) < 1$)	P
sign-balanced + γ-spectral separation	exact P (Thm. 8)
sign-balanced, bounded caps/impact/$1/\varepsilon$, inverse-poly δ	poly-time δ-approx (Prop. 6)
non-bipartite, large impact	PPAD-hard